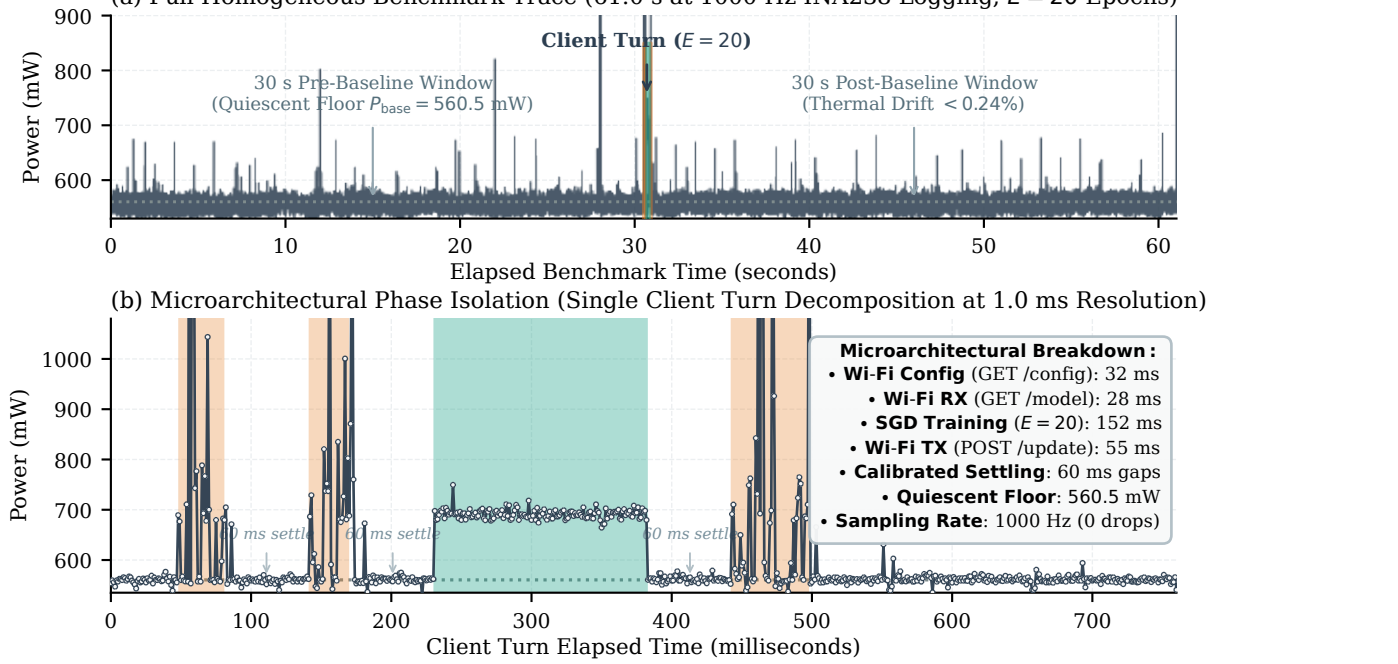

(a) Full Homogeneous Benchmark Trace (61.0 s at 1000 Hz INA238 Logging, $E = 20$ Epochs)
Power (mW)
900
800
700
600
Client Turn ($E = 20$)
30 s Pre-Baseline Window
(Quiescent Floor $P_{base} = 560.5$ mW)
30 s Post-Baseline Window
(Thermal Drift $< 0.24\%$)
0
10
20
30
40
50
60
Elapsed Benchmark Time (seconds)
(b) Microarchitectural Phase Isolation (Single Client Turn Decomposition at 1.0 ms Resolution)
Power (mW)
1000
900
800
700
600
60 ms settle
60 ms settle
60 ms settle
Microarchitectural Breakdown:
• Wi-Fi Config (GET /config): 32 ms
• Wi-Fi RX (GET /model): 28 ms
• SGD Training ($E = 20$): 152 ms
• Wi-Fi TX (POST /update): 55 ms
• Calibrated Settling: 60 ms gaps
• Quiescent Floor: 560.5 mW
• Sampling Rate: 1000 Hz (0 drops)
0
100
200
300
400
500
600
700
Client Turn Elapsed Time (milliseconds)
Instantaneous Power (1 kHz INA238)
Quiescent Baseline (560.5 mW)
Wi-Fi Communication (COMM)
On-Device SGD Training (TRAIN)